

Study guide

This guide describes the course organization and gives information on the Intended Learning Outcomes and the grading system applied.

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1) Summary

TOSK18 – Operating Systems – in ENGLISH	7.5 credits – First-cycle university education
Coordinator: Jérôme Landré	Examiner: Jérôme Landré
Digital education platform : Canvas	Jönköping University – Autumn 2025

2) Content of the course

This course is dedicated to studying an introduction to Operating Systems, both classical and real-time ones. Its purpose is to give basic knowledge of operating systems for real-time and embedded systems and how these can be configured and used by their programming interfaces.

The following topics will be covered during the course:

- Introduction to real-time systems

- Structure of operating systems, especially those with real-time capabilities
- Concurrent programming
- Scheduling of real-time tasks
- Use of operating systems (mainly Linux)

3) Intended Learning Outcomes

The intended learning outcomes of the course are found in the Course Syllabus.

ILOs	Evaluation
- Demonstrate an ability to use an operating system when developing an embedded system. - Have knowledge about and be able to explain how operating systems are constructed and work, especially when it comes to interrupts, processes, threads, and scheduling - Have knowledge about what characterizes different kinds of real-time systems - Have knowledge about and be able to explain the interaction between hardware and operating system - Have knowledge about and be able to describe some operating systems for embedded systems - Have knowledge about how interrupts, jitter, drift, etc. affects real-time systems and how the effects can be minimized - Have knowledge about how Linux operating systems are built especially regarding architecture, configuration, process handling, file systems, etc. - Demonstrate the ability to use different methods to decide if scheduling is feasible	Written Exam + Labs

4) Learning Activities

The teachers want the course to be **progressive** so that you will learn all the building blocks one after another and apply the theoretical knowledge to practical lab sessions.

a) Lectures

You will have two (or fewer) academic lectures per week that will present theoretical and practical knowledge on campus at the university.

b) Lab Sessions

Each week, you must attend a 4-hour lab session. At the lab session, you may ask the teacher for help and feedback with your labs and assignments, and you must present several labs to a teacher. The list of mandatory presentations for labs will be published on Canvas.

5) Examination Tests

The course contains two examination tests:

1. Written Examination (Inspira on campus online test): **4 credits**, graded Fail, 3, 4, or 5.
2. Labs presentation: **3.5 credits**, graded Fail/Pass.

6) Grading guide

To receive a final grade on the course, you need to pass both examination tests (Written Examination and Labs presentation).

1. Written Examination (4 credits): it is an Inspira online on campus examination that is mandatory. The questions will be graded on 100 points. The final grade will be given according to the scale:

<50	>=50	>=75	>=90
Fail	3	4	5

2. Labs presentation (3.5 credits): it consists of presenting your mandatory presentations for labs to a lab assistant. It is a pass/fail grade only.

To get the final grade and validate this course, you must pass the labs and have at least grade 3 on the Written examination.

7) Examination and re-examination

Re-exam periods are offered during the academic year (<https://ju.se/student/en/studies/academic-calendar/school-of-engineering.html>).

If you haven't passed some examination tests after the three examination opportunities (ordinary + 2 re-exam), you need to re-take those tests the next time the course runs. Note that the instructions for those tests might have changed the next time the course run, and that you should take them according to the new instructions.

The final grade can only be graded Fail(U)/3 at re-examination!

Exam week

(<https://ju.se/student/en/studies/examination/exam-schedule-jth.html>) for period 2 is:

- Week 51

Re-exam weeks for period 2 are:

- Week 7

- Weeks 23-24

The academic year 2025/2026

Autumn 2025 (2025-09-01 – 2026-01-18)

Re-exam (per 3-4) 250818-250824 v.34

Introduction week 250825-250831 v.35

Study period 1 250901-251017 v.36-42

Exam period 251018-251026 v.43

Study period 2 251027-251212 v.44-50

Exam period 251213-251221 v.51

Re-exam period (per 1+4) 260110-260118 v.3

Spring 2026 (2026-01-19 – 2026-06-07)

Study period 3 260119-260313 v.4-11

Re-exam period (per 2) 260207-260215 v.7

Exam period 260314-260322 v.12

Study period 4 260323-260522 v.13-21

Exam period 260523-260531 v.22

Graduation 260529

Re-exam period (per 1-3) 260601-260610 v.23-24

Operating Systems 7.5 HP- TOSK18

TOSK18/1801 4 HP (1801 Written examination)

Tuesday 16 December time 14:00 to 16:00

The registration will open 2025-11-16

Type of exam: Inspira – Digital exam **Anonymous**

Room: **Campus Jönköping**

Free text: Read and find the rules for exams here:

<https://ju.se/student/en/studies/examination.html>

8) Course Literature and Other Teaching Aids

The course is not based on a specific book but on many freely available resources on the Internet. However, the students **can** read two books (available at JU library) to build their own knowledge on operating systems. It is **not mandatory**:

- "Modern Operating Systems", Andrew S. Tanenbaum, Herbert Bos, ISBN: 9781292061429
- "Hard Real Time Computing Systems", Giorgio C. Buttazzo, ISBN: 9781461430193

Several online resources will be added to Canvas during the course.

9) The use of AI

In this course, it is **allowed** to use AI chatbots to get help during the lab sessions. When AI is used, students must cite the chatbot used as an APA citation (<https://guides.library.ju.se/apa-en/ai-tools>).

However in the final Inspira examination, there will be **no AI help** and **no Internet access**. Students must build their own knowledge and understanding of operating systems throughout the course to get prepared to the final Inspira examination.

10) Weeks of the course

The course will run from week 44 to week 51 in Autumn 2025 semester.

Lectures	Labs
Week 44 - Introduction & C	Lab0
Week 45 - C & Processes and Threads	Lab1
Week 46 - Linux	Lab2
Week 47 - Scheduling	Lab3
Week 49 - Deadlocks and Priority Inversion	Lab4
Week 50 - FreeRTOS	Late Lab presentations
Week 51 - Written exam	

This schedule can change during the course, an announcement will be sent to the students on Canvas before each change.